

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH223 Pharmaceutical Biochemistry

Date: 08.11.2016, Tuesday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. The primary structure of protein represents
[A] linear sequence of amino acids joined by peptide bond
[B] 3-D structure of protein
[C] helical structure of protein
[D] sub unit structure of protein
 2. Most of the enzymes are active in ____ pH range
[A] Neutral pH (6-8)
[B] Acidic pH (1-4)
[C] Alkaline pH (8-10)
[D] Entire pH range (1-14)
 3. Disulfiram used for treatment of alcoholism shows _____ type of enzyme activity.
[A] Reversible competitive inhibition
[B] Reversible non-competitive inhibition
[C] Irreversible inhibition
[D] Allosteric Inhibition
 4. Enzyme catalyses a reaction by
[A] ionic interaction
[B] increasing activation energy
[C] Providing optimum temperature
[D] lowering activation energy
 5. Which one is false about Proenzymes?
[A] They are inactive
[B] Latent Enzyme
[C] They have many isoenzymes
[D] Converted to active form when need
 6. The products generated in HMP shunt is
[A] ATP and NADPH
[B] pentose and NADPH
[C] Pentose and FADH
[D] FADH and NADPH
 7. Lipids are
[A] soluble in water and insoluble in organic solvents
[B] insoluble in water and soluble in organic solvents

8. Isomerism associated with unsaturated fatty acids is
 [D] insoluble in water and insoluble in organic solvents
 [C] soluble in water and soluble in organic solvents
 [A] Cis Trans isomerism
 [B] L-,D- isomerism
 [C] P-,R- isomerism
 [D] None
9. Cholesterol is
 [A] Yellow crystalline solid
 [B] insoluble in water and soluble in organic solvents
 [C] identified by Liebermann Buchard reaction
 [D] All of the above
10. The combined action of an aminotransferase and glutamate dehydrogenase is referred as
 [A] Transamidation
 [B] Transdeamination
 [C] Deamination
 [D] None of above
11. Mutarotation refers to change in
 [A] pH
 [B] Optical rotation
 [C] Nuclear rotation
 [D] Chemical properties
12. The carbohydrates which are sweet to taste include
 [A] Monosaccharide and oligosaccharide
 [B] Monosaccharide and polysaccharide
 [C] Polysaccharide and oligosaccharide
 [D] only monosaccharides
13. The difference between oil and fat is
 [A] Oil is solid at room temperature and fat is liquid
 [B] Fat is solid at room temperature and oil is liquid
 [C] Oil is ester of glycerol and fat is ester of long chain alcohol
 [D] Fat is ester of glycerol and oil is ester of long chain alcohol
14. If the carbon chain in a fatty acid has 8 to 14 carbons it is called
 [A] Short chain fatty acid
 [B] Medium chain fatty acid
 [C] Long chain fatty acid
 [D] Longest chain fatty acid
15. The factors involved in clotting are
 [A] Proteins
 [B] Carbohydrates
 [C] Lipids
 [D] None of the above
16. Urea is the end product of _____ metabolism.
 [A] Protein
 [B] Amino acid
 [C] Carbohydrate
 [D] None of above
17. _____ is involved in production of ATP.
 [A] Golgi Apparatus
 [B] Mitochondria
 [C] Lysosomes
 [D] All of above
18. Excess uric acid causes
 [A] Arthritis
 [B] Gout

- [C] Lesch-Nyhan syndrome
[D] All of above
19. A compound that is responsible for enhancing rate of reaction without consumed in the biochemical reaction is known as
[A] Enzymes
[B] Co-enzymes
[C] Entropy
[D] None of above
20. A programmed cell death or cell suicide is known as
[A] Cipoptosis
[B] Apoptosis
[C] Artificial death
[D] Repoptosis

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Explain the Glycolysis Pathway for metabolism of carbohydrates. **05**
B Write a note on fatty acid oxidation. **05**
C Write a note on Reversible and Irreversible Enzyme inhibition. **05**
- Q 3** Attempt any **TWO** of the following
- A** Distinguish between Glycogenesis and Glycogenolysis. **05**
B Classify the different types of Enzymes as per the IUB System. **05**
C Comment: 8 ATP molecules are produced by metabolism of 1 Glucose molecule in aerobic condition. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Explain Urea cycle in brief. **05**
B Discuss about denaturation of proteins. **05**
C Write down short note on HMP shunt. **05**
D Define and classify of lipids with examples. **05**
E Write down short note on biosynthesis of cholesterol. **05**
F Describe the different types of Enzyme Specificity. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Write down short note on salvage pathway for purine biosynthesis. **05**
B Define Biological Oxidation. Discuss about enzymes and coenzymes involved in biological oxidation. **05**
C Write a note on synthesis of catecholamines. **05**
D Write a note on pyrimidine synthesis. **05**
E Write short note on β -sheet. **05**
F Write a note on regulation of enzyme activity. **05**
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